5.

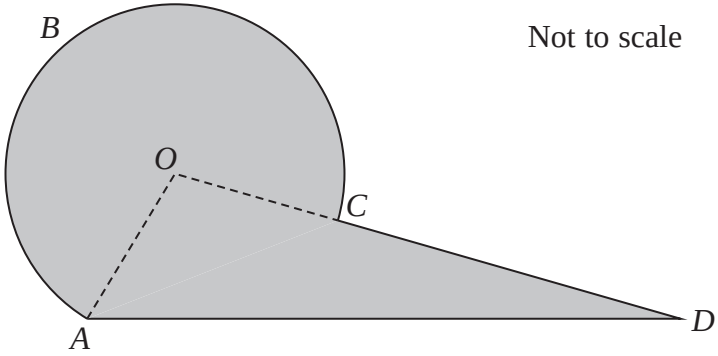


Figure 3

Figure 3 shows the plan view of a viewing platform at a tourist site.

The shape of the viewing platform consists of a sector *ABCOA* of a circle, centre *O*, joined to a triangle *AOD*.

Given that

- $OA = OC = 6\,\text{m}$
- $AD = 14\,\text{m}$
- angle $ADC = 0.43$ radians
- angle *AOD* is an obtuse angle
- *OCD* is a straight line

find

(a) the size of angle *AOD*, in radians, to 3 decimal places,

(3)

(b) the length of arc *ABC*, in metres, to one decimal place,

(2)

(c) the total area of the viewing platform, in m^2, to one decimal place.

(4)